

1 Derivation

Assume we have a first-order linear ODE, presented here in its standard (*though not strictly required*) format:

$$\frac{dy}{dx} + P(x)y = Q(x). \quad (1)$$

Our goal is to solve for $y(x)$, but how do we achieve this? To find a path forward, let's look closer at the structure of the equation itself. We can begin by defining the left-hand side as a functional acting on y :

$$T[y] = \frac{dy}{dx} + P(x)y. \quad (2)$$

Under scrutiny, one can observe that T behaves as a linear transformation, meaning it preserves both addition and scalar multiplication for functions y_1, y_2 and scalar c :

$$T[y_1 + y_2] = T[y_1] + T[y_2] \quad \text{and} \quad T[c \cdot y] = c \cdot T[y]. \quad (3)$$

Because T is a linear transformation, we can cleanly separate the operator from the function it acts upon. Rewriting the equation to make y visually distinct reveals a natural structural simplification:

$$\frac{d}{dx}y + P(x)y = Q(x) \implies \left(\frac{d}{dx} + P(x) \right) y = Q(x). \quad (4)$$

In this format, we can formally define our linear operator as $T \equiv \left(\frac{d}{dx} + P(x) \right)$, transforming our original problem into the linear system $T[y] = Q(x)$ fitting the shape $\mathbf{Ax} = \mathbf{b}$. Here, our goal is to solve for $\mathbf{x}$, which in our case would be y .

Because of the fact that for any linear system $\mathbf{Ax} = \mathbf{b}$, there always exists a homogeneous solution to $\mathbf{x}$, we know that our solution, y , must be equal to $y_h + y_p$. We can prove this by inserting our total solution into $T[y]$, giving us:

$$T[y_h + y_p] = T[y_h] + T[y_p] = 0 + Q(x) = Q(x) \quad (5)$$

Now that we know the composition of y , we can now move onto solving for each piece of the equation. To start off, we must find when $T[y] = 0$.

Solving for y will give us the homogeneous solution to our system, the set of vectors that when applied to T results in 0. Because every linear system contains some set of homogeneous solutions, we can simply set $T[y] = 0$ and solve for y . Expanding out results in:

$$\frac{dy}{dx} + P(x)y = 0 \quad \text{Which can be rearranged to be} \quad \frac{dy}{dx} = -P(x)y \quad (6)$$

Solving for y here is now quite straightforward. If we take a simple example where $P(x) = 3$ we get $y' = -3y$. A solution to this is $y = Ce^{-3x}$ which can be derived via separation of variables:

$$\frac{1}{y}dy = -3dx \implies \ln|y| = -3x + C \implies y = e^{-3x+C} = e^C e^{-3x} \implies y = ke^{-3x} \quad \text{where} \quad k = e^C \quad (7)$$

We can now define y_h as ke^{-3x} , the homogeneous solution. This leaves us with $y = ke^{-3x} + y_p$ for our solution set. The only piece missing in solving for y is our particular solution, y_p .

At this point, it may help to visualize what is actually going on when we say $T[y_h + y_p] = Q(x)$. To help bridge the gap, imagine a number-line where each point represents a function; $Q(x)$ would in turn be a single point on our line.

When we say $T[y_h + y_p]$, we are also saying $T[y_h] + T[y_p]$. If we look at just $T[y_h]$, we'll find that, in the context of our function-spaced line, this represents a line spanning the entirety of the number line. **Why?** Because in the context of our coordinate system, e^{-3x} is a single function, and it, multiplied by some constant, results in the set of all points defined by the vector:

$$k \cdot \begin{bmatrix} e^{-3x} \\ 0 \end{bmatrix} \quad (8)$$

In order for our line to pass through $Q(x)$, the particular solution, y_p , would have to take that line and perform some transformation on it to direct it toward $Q(x)$. Solving for y_p directly, however, is not ideal.

If we assume that the main role played by the particular solution is to transform the homogeneous solution in such a way that it hits the point $Q(x)$, then we can achieve the exact same thing by allowing the position of y_h to vary. This allows us to skip the search for a particular solution entirely and rely solely off the ambiguity of the constant variable in the homogeneous solution to stretch our solution over $Q(x)$.

In the context of the rest of the equation, we arrive at this:

$$T[u(x) \cdot e^{-3x}] = Q(x) \quad \text{where } u(x) \text{ is our varying constant} \quad (9)$$

$$\text{following is } \frac{d}{dx} (u(x) \cdot e^{-3x}) + 3 \cdot u(x) \cdot e^{-3x} = Q(x) \quad (10)$$

$$\text{becoming } u'e^{-3x} - 3ue^{-3x} + 3ue^{-3x} = Q(x) \quad \text{where } u' \text{ denotes } \frac{du}{dx} \quad (11)$$

$$\text{which simplifies to } u'e^{-3x} = Q(x) \quad (12)$$

And to reach our solution to $u(x)$, we simply integrate.

$$u(x) = \int Q(x)e^{3x} dx + C \quad \text{substituting back into } y: \quad y = \left(\int Q(x)e^{3x} dx + C \right) \cdot e^{-3x} \quad (13)$$

$$\text{which then becomes } y = \underbrace{\left(\int Q(x)e^{3x} dx \right) \cdot e^{-3x}}_{y_p} + \underbrace{Ce^{-3x}}_{y_h} \quad (14)$$

Finally, as a result, we now have our general solution of y to the differential equation:

$$\frac{dy}{dx} + P(x)y = Q(x) \quad \text{where } P(x) = 3 \quad (15)$$

Whether $P(x)$ is positive or negative does not affect the validity of the derivation; the sign is naturally mitigated during the application of the product rule in steps 10–11.

This completes our derivation. To conclude, we successfully derived the solution to a first-order linear ODE via critique through the lens of linear algebra and geometric thinking.